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Optimal cause marketing strategies for online platforms and third-party sellers: A spokes model analysis

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ABSTRACT

Under the advocacy of the platform, an increasing number of third-party sellers have implemented cause marketing (CM) strategies. The platform and sellers have achieved cumulative social benefits while pursuing profit. However, not all third-party sellers choose the CM strategy. To better understand the impact of online CM on pricing, profitability, and social welfare, we set up a stylized CM spokes model and analyze the impact of the unit donation amount, the number of cause marketing sellers, and consumers' warm-glow effect on the optimal CM strategy. We find that the profitability of cause-related products is not always higher than that of non cause-related products when there exists sufficient competition between cause-related and non cause-related products. It is also found that higher unit donation amounts do not necessarily lead to greater total donations and social welfare. Moreover, having more sellers choose CM does not necessarily result in a higher total donation amount. Additionally, we explore another donation form, the donation rate, to determine the best CM strategy for sellers. The results show that a fixed donation is more flexible for sellers. Overall, our findings offer new insights into how platforms and sellers can optimally implement CM strategies.

1. Introduction

Cause marketing (CM) involves donating a portion of sales from certain products to specific charitable causes [1], which can not only improve firms' profitability but also the social benefits. CM consistently plays an essential role in corporate social responsibility. Numerous prominent firms, such as Apple, Gap, and Starbucks, have implemented CM strategies. For instance, Starbucks launched the Red Product, pledging to donate a portion of the sales from those products to the Global Fund to Fight diseases, like AIDS, tuberculosis, and malaria. Similarly, IKEA promised that the entire purchase price of the soft toys designed by kids in the SAGOSKATT collection would be donated to local initiatives supporting children's rights to play, develop, and have fun. The pot brand Calphalon has launched a marketing campaign in which it will donate $5 for every frying pan sold to anti-hunger charity projects.

1.1. Characteristics of online cause marketing

Compared to traditional retail, online platforms provide a broader reach, enabling a larger number of participants and thus amplifying the social impact of CM. Additionally, consumers can engage in charitable activities through online shopping at any time and from any location. With just a few clicks, they can easily contribute to charitable causes. As a result, the benefits of online charitable activities are substantial. As online shopping continues to evolve, CM has become increasingly popular. A notable example is the *goods for good* program in Taobao, a transaction-based donation initiative that collaborates with non-profit organizations to support social causes. On this platform, sellers have the option to link their products to specific causes. When a product is connected to a social cause, it is designated as a cause marketing product, and the seller donates a portion of the retail price to the associated non-profit organization. According to the platform's policy, sellers can choose to donate either a fixed amount (e.g., ¥0.05, ¥0.1, or ¥1.0 per order) or a percentage of the retail price (e.g., 0.1 %, 0.3 %, or

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0.5 % per order), as illustrated in Fig. 1. Although each individual transaction-based donation in the program is small, the cumulative total is substantial due to the vast number of buyers and sellers participating on the platform. Transaction-based donations help generate significant social benefits by transforming small individual contributions into substantial collective impacts. In 2020, a total of 480 million consumers, 2.5 million sellers, and over 3000 non-profit organizations participated in the *goods for good* program, collectively completing over 10 billion donation transactions.

To help consumers differentiate between cause marketing products and non-cause marketing (NCM) products, CM products are marked with a sign of red heart, as illustrated in

Empirical studies have validated that some consumers are more likely to purchase socially-friendly products due to varying levels of prosociality. While some individuals are highly motivated to support social causes, others may not prioritize such considerations in their purchasing decisions as in Kraft, Valdés, and Zheng, 2018 [2]. Thus, it's interesting to discuss the impact of consumers prosociality on seller's optimal CM decisions.

In highly competitive e-commerce platforms where multiple sellers offer homogeneous products, the intensity of competition creates challenges for differentiation. To stand out, some sellers adopt CM strategies, leveraging their commitment to social responsibility as a competitive signal. While existing research has extensively explored CM strategies in monopoly or duopoly settings, such simplified market structures fail to capture the complexities of oligopoly competition, which is characteristic of e-commerce platforms. In markets with multiple sellers, the strategic interactions become more intricate, as the presence of multiple CM sellers influences both pricing and donation decisions in unique ways. Moreover, the ability of sellers to observe competitors' adoption of CM strategies significantly shapes their strategic responses. This underscores the importance of investigating how the number of CM sellers affects market outcomes. Given the growing significance of online cause marketing as a trend in corporate social responsibility, it is crucial to understand its implications within the context of highly competitive marketplaces, where the interplay between multiple CM and non-CM sellers introduces new interactions that are absent in simpler market structures.

While cause marketing (CM) has been extensively studied, several research gaps remain, particularly in the context of e-commerce platforms. First, most existing studies focus on traditional supply chains or offline markets, overlooking the distinct characteristics and challenges of CM strategies in online marketplaces. Second, the majority of the literature examines CM strategies within monopoly or duopoly settings, which do not accurately reflect the competitive landscape of e-commerce platforms, where numerous sellers operate simultaneously. Third, despite the increasing participation of sellers in CM, limited attention has been given to the role of the number of CM sellers in shaping key market outcomes, such as sellers' decisions to adopt CM. Fourth, while the warm-glow effect—a positive psychological response experienced by consumers engaging in socially responsible behavior—is well-documented, its incorporation into CM models remains underexplored, leaving a gap in understanding its influence on consumer preferences for cause-related products. Finally, there is insufficient research addressing the managerial implications of CM for third-party sellers and platforms, particularly in developing optimal donation strategies and pricing models suited to the oligopoly competition environment. These gaps highlight the necessity for a comprehensive analysis of CM strategies within competitive e-commerce platforms (Fig. 2).

Therefore, this paper explores the following issues from the perspectives of both third-party sellers and e-commerce platforms. From the seller's perspective, should sellers engage in CM, and if so, when and how should they implement CM strategies? How do unit donation amounts, consumer prosociality, and the number of CM sellers influence sellers' decision-making? From the platform's perspective, firstly, do higher unit donation amounts and more CM sellers necessarily lead to higher total donations and social welfare? How can the platform maximize the total donation amount? In this paper, we shed light on these questions.

1.2. Contribution

We propose a model of oligopoly competition that can help investigate seller's strategic implications of CM. We analyze a market consisting numerous sellers competing on an online retail platform. Consumers experience a warm-glow effect from purchasing CM products. Sellers decide whether to adopt CM and, if so, determine both the donation amount and the price for their CM products. If they choose not to adopt CM, sellers only need to decide the price of their product. We assume that each seller offers only one product on the platform. To identify the conditions under which sellers should use CM, we compare three scenarios where no sellers, some sellers, and all sellers adopt CM. Moreover, we discuss the situation where donations are given as a percentage of the price and compares it to the case where donations are fixed amounts in the main model. Finally, we investigate the impact of different commission fee rates on whether and when sellers should adopt CM. Several interesting results emerge from our research.

Firstly, we reveal that the profitability of CM products is not always higher than that of NCM products when there is sufficient competition between CM and NCM sellers. We provide conditions on whether and when to adopt CM (see Proposition 3). When consumers' prosociality is low and the number of CM sellers is small, adopting CM can help sellers

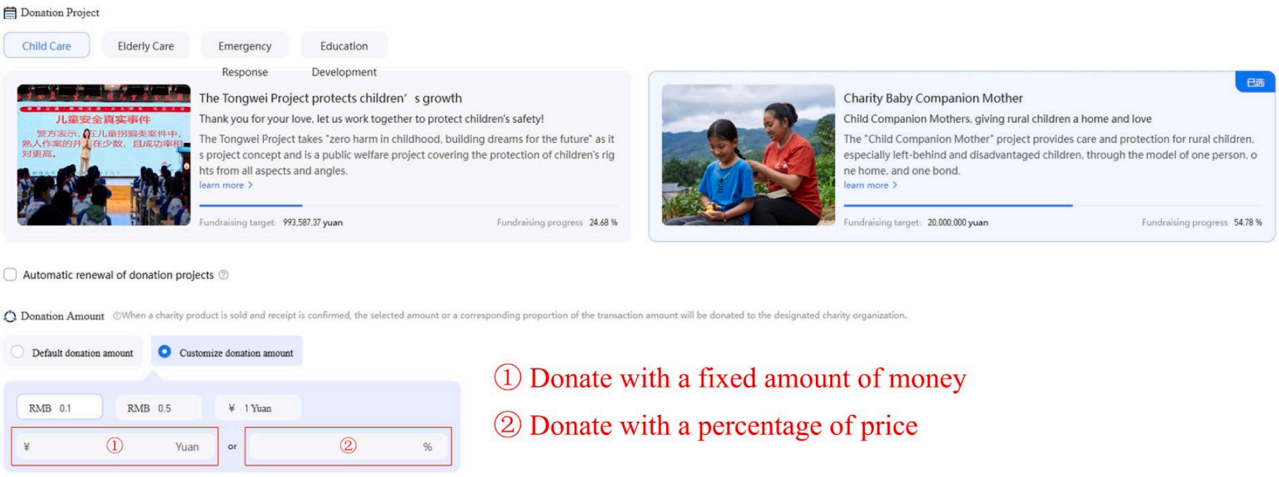


Fig. 1. Interface view of setting up a CM product on Taobao.



Fig. 2. CM and NCM products in Taobao.

distinguish themselves from others, resulting in higher profitability. However, when there are many CM sellers, the advantage of differentiation diminishes. Conversely, when consumers' prosociality is relatively high, adopting CM is more profitable due to the warm-glow effect, as the increased utility consumers gain from supporting a cause outweighs the utility lost due to the higher price of CM products.

Secondly, as the total donation is the sum of individual unit donations, one might naturally expect that higher unit donation amounts would positively impact the total donation. However, we surprisingly find that when CM and NCM sellers are in full competition, higher unit donation amounts do not necessarily lead to greater total donations and social welfare (see Proposition 4). This is because excessively increasing the unit donation amount can result in higher CM product prices, exceeding consumers' base valuation. This leads undecided consumers to choose NCM products due to the significant price difference. When the demand of CM products decreases, the total donation decreases as well. Therefore, when the decreases in the profit of CM sellers and consumer surplus are more significant than the increases in the profits of NCM sellers, the platform, and the total donation, social welfare decreases with the unit donation. Considering the trade-off between these factors, social welfare initially increases with the unit donation but eventually decreases.

Thirdly, when donations are made as a fixed percentage of the price, having more sellers choose CM does not necessarily result in a higher total donation amount. In conditions of full competition between CM and NCM sellers, and when consumer prosociality is relatively high, the total donation amount from a few sellers choosing CM is higher than when all sellers choose CM (see Proposition 6). This is because the pricing of CM products directly affects the donation amount when donations are given as a percentage. Consumers with high prosociality are willing to pay higher prices for CM products. Thus, the maximum number of CM sellers does not necessarily yield the highest total donation amount.

In addition to the above results, we compare the prices between CM and NCM products and investigate the impact of the number of CM sellers on the optimal prices (see Proposition 2). We also suggest that when the unit donation is given as a fixed percentage, sellers adopt CM when the price rate is relatively low (see Proposition 3). Furthermore, we reveal that donating a fixed amount per order is more beneficial for sellers (see Lemma 4). When donations are given as a percentage of the selling price, sellers cannot transfer the donation cost onto consumers by raising product prices, as the donation rate directly affects sellers' profitability. Sellers must balance the increased demand due to CM with the donation costs.

In summary, we make the following theoretical contributions. First, this paper constructs a spokes model incorporating the consumer warm-glow effect, thereby expanding the research on cause marketing. Second, we define the concept of actual donations made by cause-marketing sellers on e-commerce platforms. Third, we demonstrate that increasing unit donation amounts or the number of cause-marketing sellers does not necessarily result in higher total donations or improved social welfare. Fourth, we compare the fixed donation amount model with the fixed percentage of price model, offering theoretical foundations for optimal cause-marketing strategies for platforms and sellers. Finally, we validate the rationale behind certain platform policies, such as Taobao's decision to lower unit donation standards and recommend a low price rate for sellers.

The paper proceeds as follows. Section 2 summaries the related literature. Section 3 lays out our model and derives the demand functions for CM and non-CM products using the framework of a stylized spokes model. In Section 4, we first consider two special cases: when there is no cause marketing strategy and when all sellers adopt cause marketing. We then discuss a more typical scenario where only some sellers choose CM. By comparing and analyzing the equilibrium results in these three cases, we gain deeper insights into the optimal strategies for the platform and sellers. Section 5 further examines a different donation format, donating by price rate, and the impact of varying commission fees set by the platform. Section 6 concludes.

2. Literature review

Our work is mainly related to below two streams of research: (1) online price promotion strategy (2) spokes model.

2.1. Online price promotion strategy

Let's first look at the research on online price promotion strategy. With the growth of online shopping, a variety of marketing strategies have been introduced, such as cashback offers, rebates, and coupons [3–8]. In online cash-back mechanism, merchants collaborate with third-party websites by giving some affiliate fees and these websites reimburse a portion of the selling price back to consumers [5]. Wu, Ngai, and Wu [7] focus on the cash-back strategy, considering the effect of cash-back strategy on consumer surplus and social welfare. They find that the cash-back strategy leads to a typical prisoner's dilemma for merchants in which both firms lose. Lu and Moorthy [9] examine the differences between coupons and rebates, identifying the conditions under which each policy is optimal, and pinpointing that risk adversity on the part of consumers mitigates the attractiveness of rebates. Chen, Guo, and Huang [10] develop a two-period model to examine a strategic seller's conditional-rebate and pricing decisions. Yan et al. [11] investigate whether the shopping mall should provide consumers with coupons that they can obtain from the retailer in the source market and consume at the retailer in the target market. They find that when coupons are offered to consumers, the retailer in the source market has a higher price and will pay more for horizontal joint promotion. Through a choice model, Chung, Ahn, and Chun [12] demonstrate the impact of point redemption on a seller's day-to-day pricing and inventory decisions and show that a seller can benefit from allowing reward sales. Hu et al. [13] develop a game theoretic model to study the optimal decisions for the platform and seller in different advertising scenarios and cost-bearing mechanisms. Zhang and He [14] consider the advertising competition beween duopoly firms with asymmetric costs while we set a stylized model to study the optiaml cause stratgey.

2.1.1. Cause marketing

Unlike the online promotional strategies mentioned above, cause marketing (CM) is a form of corporate social responsibility in which a company donates a specified amount to a designated cause [1]. Research on CM has focused on how various factors moderate the effectiveness of CM campaigns, such as the donation amount [15,16], product type [17,18], supply chain structures [19–22] and brand popularity [23,24]. Most studies on CM use an empirical design. Koschate-Fischer, Diamantopoulos, and Oldenkotte [15] investigate the impact of donation amount on consumers' willingness to pay (WTP) by examining donation-related and cause-related customer predispositions. They find that donations to a cause significantly increase WTP in nearly all cases. Considering consumers' self-construal and cause-related social identities, Winterich and Barone [25] compare donation-based promotions with discount-based promotions. Additionally, Zhang, Cai, and Shi [26] find that price promotions positively affect CM when consumers receive a price discount. In this paper, we focus on the CM strategy in an oligopoly market. Moreover, Giebelhausen, Lawrence, and Chun [27] examine checkout charity campaigns, where the donation is made directly by consumers. Thus, in these cases, consumers exhibit socially responsible behavior instead of the company. However, in our study, third-party sellers on the e-commerce platform decide whether to join the CM and make the donations.

Numerous psychological studies have revealed that prosocial consumers often experience a positive effect, commonly referred to as the warm glow of giving, when they engage in socially responsible behavior [28–32]. Building upon this concept, Gao [19] incorporate the warm glow effect into a theoretical framework to investigate the impact of CM on the pricing, design, and distribution of cause-linked products. Their research underscores that firms can effectively employ CM campaigns to not only enhance their bottom line but also contribute to social causes. Cui et al. [22] study CM design problem within a manufacturer-retailer supply chain environment and discuss five different supply chain cases with cause effect and marketing effect. They find CM mode choice consistency or conflict between supply chain members, depending on the efficiency of expanding the market base in CM. While their focus lies on the interactions between manufacturers, firms, and consumers within a traditional supply chain, our study delves into the dynamics between e-commerce platforms, third-party sellers, and consumers. Additionally, Li, Xu, and Yang [33] delve into the pricing dynamics of cause-related products, taking into account the varying preferences of consumers. They find that consumers' differing levels of prosociality and reference recognition significantly influence pricing decisions related to cause-related products, a finding that resonates with our own research outcomes.

In reviewing the existing literature, we have identified several significant research gaps. Firstly, while many research streams have examined the various factors influencing the effectiveness of cause marketing, there is a lack of research specifically focusing on cause marketing within the context of Internet platforms. Secondly, although previous studies have extensively explored cause marketing under monopoly and duopoly competition settings, there is a notable absence of research addressing its implications under oligopoly competition, which more closely mirrors the dynamics of the actual e-commerce landscape. Thirdly, there is a gap in research concerning cause marketing from the perspective of the number of CM sellers involved. These critical research gaps serve as the primary motivation for our study, as we aim to address these voids. Our paper contributes to the literature by filling these gaps. To distinguish previous similar literature from our work, we want to point out that Xu and Li [34] discuss an e-tailer's demand information-sharing strategy in the context of the supplier's CM decision within the Internet environment, their focus primarily revolves around information-sharing problems, which differs from the scope of our research. Lee and Charles [35] examine the interplay between retailers' ethics and promotional strategies, offering insights to e-commerce practitioners regarding information management. In contrast to their focus, our study centers on three pivotal factors in Internet cause marketing: unit donation amount, consumer prosociality, and the number of CM sellers. Furthermore, we offer managerial insights to both e-commerce platform and third-party sellers.

2.2. Spokes model

In addition to the aforementioned research, this paper is also related to the Spokes model. Initially proposed by Chen, Moorthy, and Zhang [3], the Spokes model represents a further advancement of the Hotelling model [36] and the Salop model [37], and has since become a widely accepted spatial competition model. The Spokes model is particularly well-suited for analyzing cause marketing strategies in markets with multiple competing sellers, as it differs from other models like Hotelling and Salop by eliminating the need for spatial position adjustments when the market size changes. The Hotelling model, primarily focused on duopoly competition, doesn't accommodate markets with more than two competitors effectively. Meanwhile, the Salop model does address competition among multiple sellers but requires that their positions be adjusted in response to changes in market size, which can complicate analysis in dynamic, real-world scenarios. The Spokes model offers a simpler structure that maintains fixed competitive positions, making it particularly well-suited for examining how multiple sellers simultaneously approach cause marketing, without the added complexity of adjusting spatial positioning. This feature aligns well with the objectives of this study, which aims to understand the strategic interactions among sellers utilizing cause marketing strategies. As such, the Spokes model is chosen as the framework for this analysis, as it facilitates a clearer examination of competitive dynamics in the context of cause marketing.

The Spokes model has been widely used in studies of oligopoly

competition. Chen and Riordan [38] analyzed the impact of n feasible firms on equilibrium prices and profits, laying the groundwork for further exploration of spokes model. Building on this work, Caminal and Granero [39] examined the optimal product selection problem in scenarios where firms offer multiple types of products, contrasting with Chen and Riordan [38], where firms offer only a single product. Using the Spokes model, Amaldoss and He [40] extended its application to explore pricing issues for benchmark and non-benchmark products in an oligopoly market, incorporating consumer loss aversion into their analysis, which adds a novel behavioral dimension to the model. Additionally, Lei et al. [41] utilized the model to examine pricing patterns and competition strategies under the opaque pricing of residual inventory distribution models. Furthermore, Sharma and Mehra [42] on the other hand, utilized a three-firm Spokes model to analyze the impact of platform entry into external product markets, specifically focusing on how the quality of external products influences the platform's advertising business profits. Distinct from the above studies, this paper integrates consumer warm-glow utility into the Spokes model, providing a new perspective on cause marketing strategies. By accounting for the additional positive utility consumers gain from purchasing cause-related products, this study contributes a unique angle to the literature on strategic interactions in competitive markets.

3. Model description

Consider a marketplace platform, such as Taobao or T-mall, hosting T third-party sellers offering an undifferentiated product, where $T \geq 2$. Among these sellers, those adopting a cause-related marketing (CM) strategy are referred to as CM sellers, and their products are CM products. The total number of CM sellers is M, indexed by $i, j = 1, 2, \ldots, m$. Sellers who do not join the CM strategy are called NCM sellers, offering NCM products. The total number of NCM sellers is N, indexed by $z, s = 1, 2, \ldots, n$. Each seller is assumed to sell only one product. For simplicity and without loss of generality, we assume that the market is fully covered, meaning all available products are sold. Thus, the total number of third-party sellers satisfies $T = M + N$.

In the plane of the spokes model, as illustrated in Fig. 3, each seller is represented by a point that is the origin of a line of length 1 /2. The other end of the line is called its terminal. For variety i, its associated line is called l_i, and the terminals of all the lines meet at one point, called the center [38]. This forms a radial network of lines (spokes network) connected at the center, representing the preference space. Fig. 3 illustrates a market where exist $T = 8$ products, and $M = 3$ of them are CM products, $N = 5$ of them are NCM products. Consumers typically make a purchase decision within a small consideration set. In keeping with prior literature[38,40,43], we assume that consumers ultimately choose between at most two products. This assumption is based on three key reasons: Firstly, in practice, consumers rarely compare all T sellers before making a purchase decision due to comparison costs, search costs, and cognitive limitations. As a result, consideration sets are typically very small, and consumers often narrow their final decision to just 2–3 options. Secondly, this assumption allows for the derivation of a pure strategy Nash equilibrium. Thirdly, limiting the choice set simplifies the analytical process and helps focus on key market decisions.

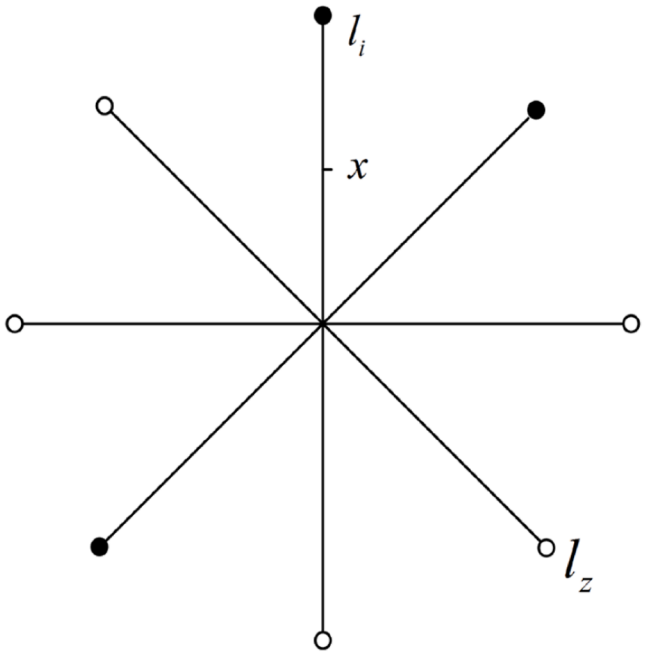


Fig. 3. A spokes model with $\boldsymbol{T} = 8$, $\boldsymbol{M} = 3$ and $\boldsymbol{N} = 5$.

Moreover, we assume that a consumer's first-preferred product is the local product corresponding to the spoke where the consumer resides. The second-preferred product can be any nonlocal product, each with an equal probability of $\frac{1}{T-1}$. For example, if a consumer x resides in l_i, their first-preferred product is i. Any of the nonlocal products may serve as their second-preferred option, with equal likelihood. $\frac{1}{T-1}$.

The utility that consumers derive from purchasing the local product is given by:

$$U = v - p - tx + \delta, \tag{1}$$

where v is the consumer's base valuation, p is the product's purchase price, t represents the consumer's sensitivity to product characteristics or the traveling cost. Many empirical studies have demonstrated that consumers derive additional positive utility when purchasing cause-linked products [28]. Building on this, we assume that consumers experience a warm-glow effect when they buy a CM product [15,19]. The magnitude of this effect is denoted by δ ($\delta \geq 0$), representing the additional utility consumers gain from the cause-related aspect of the product. For products that are not linked to a cause (NCM products), the warm-glow effect is zero, i.e., $\delta = 0$.

Sellers need to decide whether to adopt CM. If they choose to adopt CM, they must determine the product price p and the unit donation amount d, where $d < p$. If they choose not to adopt CM, they only need to determine the product price p, and $d = 0$. The third-party seller has to pay a commission fee to the online platform, which is a fraction $r \in (0, 1)$ of the sales revenue. This fee structure is commonly used by many online platforms, such as Taobao, Amazon, and eBay. Since the shopping platform does not change the commission fee frequently and the rate is fixed in the short term, without loss of generality, we assume that the commission rate is exogenous. We further discuss the impact of different commission fee rates in the extension section and demonstrate that changes in the commission rate will not fundamentally alter the main results. Therefore, the profit functions of the online platform and third-party sellers are

$$\pi_0 = r\left(\sum_{i=1}^{m} p_i Q^{CM} + \sum_{z=1}^{n} p_z Q^{NCM}\right), i = 1, 2, \ldots, m; z = 1, 2, \ldots, n, \tag{2}$$

$$\pi_i^{CM} = [(1-r)p_i - d]Q_i^{CM}, i = 1, 2, \ldots, m, \tag{3}$$

$$\pi_z^{N-CM} = (1-r)p_z Q_z^{N-CM}, z = 1, 2, \ldots, n, \tag{4}$$

where π_i^{CM} and π_z^{N-CM} are the profit of CM sellers and NCM sellers respectively.

We formulate a two-stage game to analyze the decision-making processes of third-party sellers and consumers. In the first stage, given the commission rate r set by the platform, the seller announces the product price p and decides whether to adopt the CM strategy with a unit donation amount d. In the second stage, consumers make purchases if their utility U is greater than zero. Backward induction is applied

throughout this study. The notations used in this paper are listed in Table 1.

3.1. Segmentation of a seller with cm

Due to market uncertainty and the fact that not every seller prefers the CM model, we derive the demand for both CM and NCM products. In Fig. 4, we illustrate the consumer segments where a given CM product i is the most preferred product in segments a and b, while for consumers in segments c and d, it is their second-preferred product. And then we present the demand function for each segment of consumers.

Depending on the consumers' preferrence, for example, consumers in segment a decide on whether to buy the product from seller i or seller j. The location of the marginal consumer who is indifferent between the two sellers is given by $v - p_i - tx + \delta = v - p_j - t(1-x) + \delta$. Note that for any consumer located on l_i, seller j is her second-preferred seller with probability $1/(T-1)$, where $j \in \{1, \cdots, m+n\}$ and $j \neq i$. Since consumers are evenly distributed across all spokes, the density of such consumers is $2/T$. Therefore, the demand from these consumers can be described as follows.

Segment a: Consumers in the segment a need to decide whether to purchase the product from i or j, both of whom are cause marketing product, where $j \in \{1, \cdots, m\}$ and $j \neq i$. Such a consumer is indifferent between varity i and j if $v - p_i - tx + \delta = v - p_j - t(1-x) + \delta$. The demand from these consumers is

$$\frac{2}{T}\frac{1}{T-1}\sum_{j=1,j\neq i}^{m}\max\left\{\min\left\{\frac{1}{2}-\frac{p_i-p_j}{2t},1\right\},0\right\}. \tag{5}$$

Segment b: Consumers in this segment consider NCM products to be their second-preferred, and they need to choose between CM products and NCM products. The indifferent point in this segment is $v - p_i - tx + \delta = v - p_z - t(1-x)$, where $z \in \{1, \cdots, n\}$. Hence, the demand from this group can be written as

$$\frac{2}{T}\frac{1}{T-1}\sum_{z=1}^{n}\max\left\{\min\left\{\frac{1}{2}-\frac{p_i-p_z-\delta}{2t},1\right\},0\right\}. \tag{6}$$

Segment c: Consumers need to decide whether to buy the NCM product or CM product. The indifferent point in this segment is $v - p_i - tx + \delta = v - p_z - t(1-x)$, where $z \in \{1, \cdots, n\}$. Hence, the demand function is the same as segment b. We named this kind of phenomenon mirror image, which means the one side situation is the same as a model in the mirrors.

Table 1
Summary of Notations.

Notation	Description
v	Base value of the product
p	Price of the product
x	Consumer's location
t	Consumer's sensitivity to product characteristics
δ	Consumer's warm-glow effect
d	The fix unit donation amount given by the seller
U	Utility of purchasing a product
r	Commission fee rate given by the platform
ϵ	Price rate when the donation is made as a fixed percentage of the retail price
ω	Total donation amount in the platform
q	Demand for products
i,j	Index of cause marketing sellers
z,j	Index of non-cause marketing sellers
m	The mth cause marketing product
n	The nth non-cause marketing product
M	Number of third-party sellers who are adopting cause marketing
N	Number of third-party sellers who are not adopting cause marketing
T	Total amount of third-party sellers on a platform
CS	Consumer surplus
SW	Social welfare

Segment d: For the mirror image, the demand function is the same as segment a.

We assumed that the demand function of CM product could be written as

$$q^{CM} = \frac{2}{T}\frac{1}{T-1}\left(\sum_{j=1,j\neq i}^{m}\max\left\{\min\left\{\frac{1}{2}-\frac{p_i-p_j}{2t},1\right\},0\right\} + \sum_{z=1}^{n}\max\left\{\min\left\{\frac{1}{2}-\frac{p_i-p_z-\delta}{2t},1\right\},0\right\}\right). \tag{7}$$

3.2. Segmentation of a seller with n-cm

In this section, a given NCM product z is the most preferred product in segment N-a and N-b, whereas for consumers in segment N-c and N-d, it is their second-preferred product. We illustrate the consumer segments in Fig. 5 and derive the demand function for each segment in the following part.

Segment N-a: For consumers in this segment, product i is their second-preferred product. The indifferent point in this segment is $v - p_z - tx = v - p_i - t(1-x) + \delta$, where $i \in \{1, \cdots, m\}$. Hence, the demand from these consumers is

$$\frac{2}{T}\frac{1}{T-1}\sum_{i=1}^{m}\max\left\{\min\left\{\frac{1}{2}-\frac{p_z-p_i+\delta}{2t},1\right\},0\right\}. \tag{8}$$

Segment N-b: Consumers need to choose between two NCM products. The indifferent point in this segment is $v - p_z - tx = v - p_s - t(1-x)$, where $s \in \{1, \cdots, n\}$, and $s \neq z$. Hence, the demand for z is

$$\frac{2}{T}\frac{1}{T-1}\sum_{s=1,s\neq z}^{n}\max\left\{\min\left\{\frac{1}{2}-\frac{p_z-p_s}{2t},1\right\},0\right\}. \tag{9}$$

Segment N-c(N-d): For the mirror image, the demand function is the same as segment N-b(N-a).

We assume that, for these consumers, the demand function of NCM products z can be written as

$$q^{NCM} = \frac{2}{T}\frac{1}{T-1}\left(\sum_{i=1}^{m}\max\left\{\min\left\{\frac{1}{2}-\frac{p_z-p_i+\delta}{2t},1\right\},0\right\} + \sum_{s=1,s\neq z}^{n}\max\left\{\min\left\{\frac{1}{2}-\frac{p_z-p_s}{2t},1\right\},0\right\}\right) \tag{10}$$

4. Analysis

4.1. No cause marketing (N—CM)

In this section, we assume the CM doesn't exist and none of the sellers can make donations. Therefore, all the sellers are NCM sellers, that is $T = N(N \geq 2)$ and $M = 0$. The demand function is

$$Q_z^{NCM} = \frac{2}{T}\frac{1}{T-1}\sum_{s=1,s\neq z}^{n}\left(\frac{1}{2}-\frac{p_z-p_s}{2t}\right) \tag{11}$$

Lemma 1. *When all sellers do not have cause marketing strategy, the equilibrium prices and profits of NCM products are given by* $p_z^* = p_s^* = t$; $\pi_z^* = \pi_s^* = \frac{t(1-r)}{T}$.

Lemma 1 describes the equilibrium outcome in a market with no cause marketing (CM) sellers, where the behavior of consumers, sellers, and the platform mirrors that of a standard Hotelling model. In this scenario, consumers choose between two non-cause-related (NCM) products, and the NCM sellers engage in pairwise Hotelling-style duopoly competition. This leads to an equilibrium where product prices are equal and there is complete substitution between the products.

In this equilibrium, the market is evenly divided into T parts, with each seller achieving the same profit level. However, as the number of

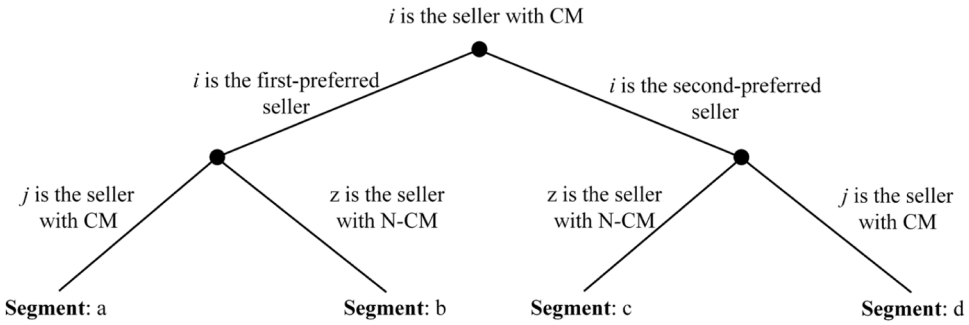


Fig. 4. Consumers segmentation with CM.

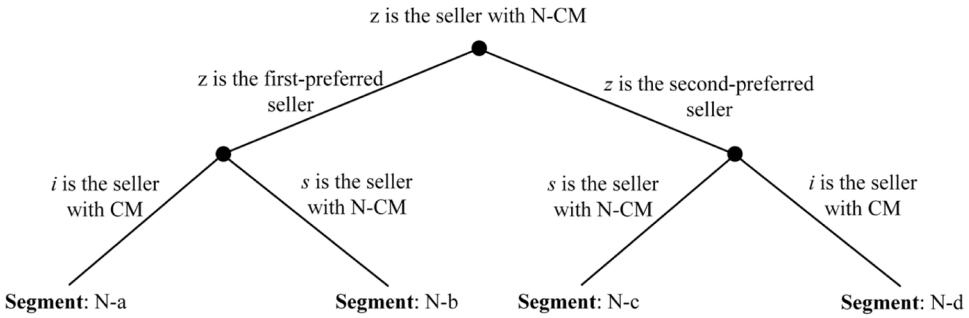


Fig. 5. Consumers segmentation with N—CM.

sellers offering a homogeneous product increases, competition intensifies. This increased competition arises because a higher number of competing sellers gives consumers more choices, which in turn lowers product prices and reduces the profit levels for sellers

This result can be understood through competition theory in economics. As the number of sellers in the market increases, competition intensifies. This heightened competition forces sellers to reduce their prices to attract more consumers and boost sales volume. Consequently, the increased competition leads to a decrease in product prices. As a result, the rising number of sellers and the corresponding increase in competition cause a decline in the profit levels of individual sellers.

Hence, Lemma 1 demonstrates that, in the absence of CM sellers, the equilibrium outcome in the market resembles that an increase in the number of sellers leads to more intense competition and a decrease in profit levels. Detailed calculations are provided in the **Appendix A**.

4.2. Full cause marketing (F-CM)

Now we consider a special case where all sellers adopt the CM strategy, and all products in the market are cause-related. In this scenario, for every purchase, a portion of the payment is donated to a specific charity project. This case is the opposite of the previous one, where $T = M (M \geq 2)$ and $N = 0$. Consequently, the demand function is

$$Q_i^{CM} = \frac{2}{T}\frac{1}{T-1}\sum_{j=1 j\neq i}^{m}\left(\frac{1}{2}-\frac{p_i - p_j}{2t}\right) \tag{12}$$

Lemma 2. *When all sellers adopt the CM strategy, the equilibrium prices, profits of CM products and total donation are given by* $\boldsymbol{p_i}^* = \boldsymbol{p_j}^* = \boldsymbol{t} + \frac{\boldsymbol{d}}{\boldsymbol{1-r}}$, $\boldsymbol{\pi_i^*} = \boldsymbol{\pi_j^*} = \frac{\boldsymbol{t(1-r)}}{\boldsymbol{T}}$ *and* $\boldsymbol{d}$.

Lemma 2 presents the equilibrium results when all sellers adopt the CM strategy. Similar to Lemma 1, when all sellers choose to adopt the CM strategy, they engage in Hotelling competition. In this scenario, the equilibrium prices and profits remain consistent for all sellers. When all sellers adopt the CM strategy, the optimal price is influenced by the rate of unit donation to commission fee and the consumer's sensitivity to product characteristics. Compared to the results in Lemma 1, sellers adopting the CM strategy set a higher price for the CM product.

When sellers make donations as a fixed unit donation amount, they effectively transfer the donation cost to consumers by raising the price. Compared with the optimal prices we get in N—CM, the additional price $\frac{d}{1-r}$ indicates how much sellers raise the price to cover the donation cost. Hence, we denoted $\frac{d}{1-r}$ as the actual donation amount given by consumers. In contrast, the donation amount d promised by the seller, denoted as, can be understood as the nominal donation amount shown to consumers.

An interesting finding from Lemma 2 is that the actual donation amount $\frac{d}{1-r}$ is higher than the nominal donation amount promised by sellers. This discrepancy arises because sellers need to pay the commission fee to the platform, a factor that differentiates this cause marketing strategy from those applications in traditional channels.

One might wonder why the extra positive utility, the warm-glow effect δ, from purchasing CM products vanishes in the outcome. From a psychological perspective, besides charitable consciousness, the scarcity and uniqueness of cause-related goods are crucial factors that motivate consumers to make purchases [44,45]. When all products in the market are cause-related, the extra positive effect derived by consumers diminishes or even disappears because CM products are no longer unique. This explains why δ does not appear in the equilibrium results in this scenario.

Proposition 1. *(1)* $\boldsymbol{\pi_0}^{\boldsymbol{F-CM}} > \boldsymbol{\pi_0}^{\boldsymbol{N-CM}}$; *(2) When* $\boldsymbol{\delta} > \frac{\boldsymbol{d}}{\boldsymbol{1-r}}$, $\boldsymbol{CS}^{\boldsymbol{F-CM}} > \boldsymbol{CS}^{\boldsymbol{N-CM}}$; *(3) When* $\boldsymbol{\delta} > \frac{\boldsymbol{r(M-1)d}}{\boldsymbol{(1-r)M}}$, $\boldsymbol{SW}^{\boldsymbol{F-CM}} > \boldsymbol{SW}^{\boldsymbol{N-CM}}$.

Proposition 1 compares two special scenarios: when none of the sellers have cause marketing (N—CM) and when all sellers have cause marketing (F-CM), in terms of platform profit, consumer surplus, and social welfare.

Specifically, Proposition 1**(1)** indicates that when all the sellers adopt CM, the platform achieves higher profitability compared to the N—CM scenario. This demonstrates that adopting a cause marketing strategy can always effectively enhance the platform's economic benefits. Therefore, Proposition 1**(1)** provides theoretical support for similar

charitable marketing activities launched by platforms like Taobao and Tmall's *Goods for good* program, and Meituan's *Green Mountain Plan*.

Social welfare comprises consumer surplus, third-party seller and platform profits, and the total donation amount. Proposition 1**(2)** indicates that when consumer's warm glow effect (δ) is higher than the actual donation amount ($\frac{d}{1-r}$), consumer surplus is higher in the F-CM scenario compared to the N—CM scenario. Otherwise, consumer surplus in F-CM is lower than that in N—CM.

Proposition 1**(3)** suggests that when the consumer's warm glow effect δ is higher than $\frac{r(M-1)d}{(1-r)M}$, social welfare is higher than that in the F-CM. This threshold, $\frac{r(M-1)d}{(1-r)M}$, is determined by the number of CM sellers (M), the unit donation amount (d) and the commission fee rate (r). When the nominal donation amount d increases, the threshold also rises, suggesting that a lower donation amount is more likely to positively impact social welfare. Besides, when $\frac{r(M-1)d}{(1-r)M} < \delta < \frac{d}{1-r}$, social welfare in the F-CM is higher, though consumer surplus is lower than in the N—CM. This implies that if consumers have relatively high prosociality, adopting CM universally (where all sellers participate in CM) maximizes social welfare. Conversely, if consumers' prosociality level is not relatively high ($\delta < \frac{r(M-1)d}{(1-r)M}$), not adopting CM is preferable for enhancing social welfare.

The comparison of consumer surplus and social welfare in N—CM and F-CM is demonstrated in Fig. 6 with the total number of sellers $T = 60$, consumers' sensitivity to product characteristic $t = 1$, platform commission rate $r = 0.15$, and basic consumer utility $v = 0$. Region I indicates that when consumers' warm glow effect is very high ($\delta > \frac{d}{1-r}$), having all sellers adopt CM can always bring higher consumer surplus and social welfare ($CS^{F-CM} > CS^{N-CM}$, $SW^{F-CM} > SW^{N-CM}$); Region II shows that when the warm glow effect is relatively high ($\frac{r(M-1)d}{(1-r)M} < \delta < \frac{d}{1-r}$), the consumer surplus in F-CM is lower than in N—CM, but higher social welfare can be achieved ($CS^{F-CM} < CS^{N-CM}$ and $SW^{F-CM} > SW^{N-CM}$). In this Region, the platform can achieve higher profit with CM. In other words, in F-CM, when $\frac{r(M-1)d}{(1-r)M} < \delta < \frac{d}{1-r}$, consumers sacrifice their surplus value but contribute to a higher social welfare. Region III demonstrates that when consumer' prosociality is relatively low ($\delta < \frac{r(M-1)d}{(1-r)M}$), it leads to lower consumer surplus and social welfare ($CS^{F-CM} < CS^{N-CM}$ and $SW^{F-CM} < SW^{N-CM}$). Hence, it is evident that when consumers' prosociality is relatively high ($\delta > \frac{r(M-1)d}{(1-r)M}$), having all sellers adopt CM will certainly lead to higher social welfare.

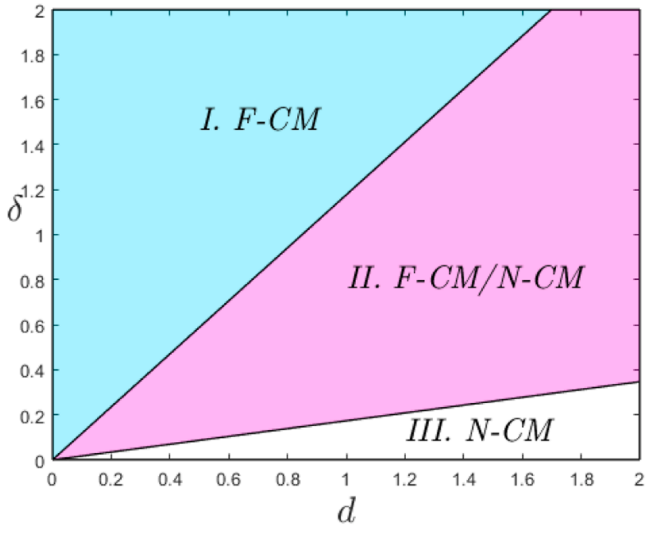


Fig. 6. The comparison of consumer surplus and social welfare in N—CM and F-CM.

4.3. Partial cause marketing (P-CM)

In this session, we will discuss a more normal case where only partial sellers choose CM. Thus, we assume $T = M + N$ and $M \geq 2, N \geq 2$. When $p_i - p_j > t$, the price gap becomes so large that leads to monopolization, which violates the assumption of competition among the T sellers. Therefore, we will only discuss the cases where $|p_i - p_j| < t$.

As shown in Fig. 4 and Fig. 5, competition arises among CM sellers (e.g., i and j), among NCM sellers (e.g., z and s), and between CM and NCM sellers (i and z), depending on different consumer types. When two sellers of the same type (CM or N—CM) compete, they evenly split the market at equilibrium, similar to results from the Hotelling model. However, differing tradeoffs between price gaps and the warm glow effect create unique competition cases and equilibrium outcomes. This highlights the importance of analyzing various price and warm-glow effect conditions. Two scenarios emerge when a CM seller competes with an NCM seller. We denote the MM case (CM sellers monopolize the market) when $-t < p_i - p_j < \delta - t$, and the MN case (intense competition between CM and NCM sellers) as when $\delta - t < p_i - p_j < \delta + t$. Next, let's look at the demand functions in the two cases respectively.

(1) CM sellers monopolize the market (MM)

When the price gap between CM and NCM sellers is small, consumers who are indecisive between CM and NCM sellers (a2, b2, N-a1, N-b1 type consumers) may choose the CM seller due to the additional positive utility gained from purchasing CM products. Consequently, the CM seller could monopolize the market. The demand function can be expressed as follows:

$$Q_i^{CM} = \frac{2}{T}\frac{1}{T-1}\sum_{j=1,j\neq i}^{m}\left(\frac{1}{2}-\frac{p_i-p_j}{2t}\right) \tag{13}$$

$$Q_z^{NCM} = \frac{2}{T}\frac{1}{T-1}\left(\sum_{s=1,s\neq z}^{n}\left(\frac{1}{2}-\frac{p_z-p_s}{2t}\right)+M\right) \tag{14}$$

(2) Competition between the CM and NCM sellers (MN)

Due to differences in consumers' prosociality, when the price gap between CM and NCM products is large, there will be sufficient competition between CM and NCM sellers. The demand functions can be expressed as follows:

$$Q_i^{CM} = \frac{2}{T}\frac{1}{T-1}\left(\sum_{j=1,j\neq i}^{m}\left(\frac{1}{2}-\frac{p_i-p_j}{2t}\right)+\sum_{z=1}^{n}\left(\frac{1}{2}-\frac{p_i-p_z-\delta}{2t}\right)\right) \tag{15}$$

$$Q_z^{NCM} = \frac{2}{T}\frac{1}{T-1}\left(\sum_{s=1,s\neq z}^{n}\left(\frac{1}{2}-\frac{p_z-p_s}{2t}\right)+\sum_{i=1}^{m}\left(\frac{1}{2}+\frac{p_i-p_z-\delta}{2t}\right)\right) \tag{16}$$

We substitute Q_i^{CM} and Q_z^{NCM} into the profit functions respectively, and maximize π_i^{CM} and π_z^{NCM}, then we obtain the equilibrium prices, and all the calculations are shown in **Appendix A**.

Lemma 3. *When partial sellers have cause marketing strategy,*

(1) *In MM, the equilibrium prices, profits of CM products and total donation are* $p_i = t + \frac{2Nt}{M-1}$, $\pi_i = \frac{(1-r)(M+2N-1)^2 t}{T(T-1)(M-1)}$ *and* $\omega = \frac{dM(M+2N-1)}{T(T-1)}$; *the equilibrium prices and profits of NCM products are* $p_z = t$ *and* $\pi_z = \frac{(1-r)(N-1)t}{T(T-1)}$;

(2) *In MN, the equilibrium prices, profits of CM products and the total donation are* $p_i = t + \frac{N\delta}{2T-1}$, $\pi_i = \frac{\left(\left(\left(t+\frac{\delta}{2}\right)(r-1)+\frac{d}{2}\right)N+t(r-1)\left(M-\frac{1}{2}\right)\right)^2}{T(1-r)\left(T-\frac{1}{2}\right)^2 t}$

and $\omega = \frac{M\left(\left(\left(t+\frac{\delta}{2}\right)(r-1)+\frac{d}{2}\right)N+t(r-1)\left(M-\frac{1}{2}\right)\right)d}{T(r-1)\left(T-\frac{1}{2}\right)t}$; the equilibrium prices and profits of NCM products are $p_z = t - \frac{((r-1)\delta+d)M}{(r-1)(2T-1)}$ and $\pi_z = \frac{\left(((1-r)(2t-\delta)+d)M+2t(1-r)\left(N-\frac{1}{2}\right)\right)^2}{(T(1-r)(2T-1))^2 t}$.

Proposition 2. *The optimal price of CM product is always higher than the NCM product. In MM, the price of CM products decreases with M; In MN, the price of CM products increases (decreases) with M when consumers' prosociality level* $\delta < \frac{d}{1-r}\left(\delta > \frac{d}{1-r}\right)$.

Proposition 2 compares the optimal prices of CM and non-cause-related (NCM) products, highlighting the impact of the number of CM sellers on optimal pricing. When sellers choose CM, they incur additional donation costs. To pass these costs onto consumers, the price of CM products is always higher than that of NCM products. Consumers are willing to pay this higher price because they gain additional warm-glow utility from their purchases. Thus, CM sellers can capture more consumer surplus by raising the prices.

As more CM sellers enter the market, increased competition lowers the price of CM products, reducing the price gap between CM and NCM products. However, counterintuitively, when there is competition between CM and NCM sellers and consumers' prosociality is low (i.e., $\delta < \frac{d}{1-r}$), the entry of more CM sellers leads to higher prices for both CM and NCM products. Notably, the prices of CM products rise faster than those of NCM products. This can be explained as follows: with more sellers choosing CM, the original demand for CM products is split among more sellers. Given the low prosociality of consumers, the overall demand for CM products decreases. To compensate for the loss in profit due to reduced demand, CM sellers raise their prices to maintain profitability.

Conversely, if consumers' prosociality is high, the entry of more CM sellers leads to a decrease in the prices of both CM and NCM products, thereby narrowing the price gap. This price decline can be attributed to CM sellers combining CM with traditional price discount promotions to attract more consumers. For example, the Amazon platform has offered price discounts on products participating in fundraising activities for the Red Cross, and the clothing company Gap has implemented a 30% price discount policy for its CM products. Proposition 2 provide a reasonable and sufficient explanation for such phenomena.

Previous literature has also reached conclusions similar to Proposition 2. Gao [19] believes that firms should reduce product price if and only if the size of prosocial segment is not very large. Krishna and Rajan [46] argue that if a single firm in the market links only one product to a cause, it can raise prices on both products and earn a higher profit. However, they only considered the case of one or two sellers and did not account for the impact of varying numbers of CM sellers in the same market on pricing. This paper expands on their conclusions by exploring how the number of CM sellers affects pricing strategies in a competitive market. As a result, it provides a more comprehensive theoretical reference for firms implementing CM, representing a significant advancement and refinement of previous literature.

Proposition 3. *In MM, the profit of CM sellers is higher than those of NCM sellers. In MN, when* $\boldsymbol{\delta} < \frac{(\boldsymbol{d}-4(1-\boldsymbol{r})\boldsymbol{t})\boldsymbol{N}+2(1-\boldsymbol{r})\boldsymbol{t}}{\boldsymbol{N}(1-\boldsymbol{r})} < \frac{\boldsymbol{d}}{1-\boldsymbol{r}}$ *and* $\boldsymbol{M} < \boldsymbol{M}_0$ $(\boldsymbol{M} > \boldsymbol{M}_0)$, *the profit of CM sellers is higher (lower) than those of NCM sellers. When* $\boldsymbol{\delta} > \frac{\boldsymbol{d}}{1-\boldsymbol{r}}$, *the profit of CM sellers is higher than those of NCM sellers, where* $\boldsymbol{M}_0 = \frac{2(1-\boldsymbol{r})(1-2\boldsymbol{N})\boldsymbol{t}+\boldsymbol{N}((\boldsymbol{r}-1)\boldsymbol{\delta}+\boldsymbol{d})}{(1-\boldsymbol{r})(4\boldsymbol{t}-\boldsymbol{\delta})+\boldsymbol{d}}$.

Proposition 3 compares the profitability of sellers with and without CM and offers insights on whether and when to adopt CM, highlighting the interaction among consumers' prosociality, the number of CM sellers, and profitability.

According to Proposition 1, the price of CM products is always higher than that of NCM products. When CM sellers monopolize the market, consumers choosing between CM and NCM products will tend to opt for CM products due to the warm-glow effect. At this point, the demand and price for CM products are higher than those for NCM products, resulting in CM sellers always having higher profits than NCM sellers. Therefore, a seller should adopt CM when most other sellers are using it.

However, Proposition 3 indicates that the profitability of CM products is not always higher than that of NCM products when there is sufficient competition between CM and NCM sellers. When consumers' prosociality is low and the number of CM sellers is small, adopting CM can help sellers distinguish themselves from others, resulting in higher profitability. Conversely, when consumers' prosociality is low and there are many sellers adopting CM, the advantages of differentiation weaken. With a relatively low level of prosociality, consumers are less likely to buy higher-priced CM products. Therefore, when consumers' prosociality level is low, sellers should adopt CM when there are few CM sellers, but avoid it when there are many CM sellers.

On the other hand, when consumers' prosociality is relatively high, sellers should adopt CM to improve their demand. In this scenario, the utility increase due to the warm-glow effect outweighs the decrease caused by the higher price. Thus, sellers can benefit from increased demand and enhanced consumer loyalty by leveraging the positive perceptions associated with cause marketing.

This proposition provides a basis for deciding whether and when sellers should adopt CM strategies. Before making decisions on pricing and whether to adopt CM, sellers should consider the prosociality of their target customer group and the current number of CM sellers. If the target customer group has high prosociality, sellers should have cause marketing strategy. For example, empirical studies have shown that women generally exhibit higher prosociality than men. Therefore, sellers whose primary target customers are women are more suitable for cause marketing. Moreover, if a seller opts for cause marketing, the pricing of CM products should not be excessively high. A relatively small price gap compared to NCM products can actually result in higher profitability.

Proposition 4. *(1) In MM, the total donation increases with the unit donation* $\boldsymbol{d}$. *In MN, if* $0 < \boldsymbol{d} < \widetilde{\boldsymbol{d}}\left(\boldsymbol{d} > \widetilde{\boldsymbol{d}}\right)$, *total donation increases (decreases) with* $\boldsymbol{d}$, *where* $\widetilde{\boldsymbol{d}} = \frac{(1-\boldsymbol{r})((\boldsymbol{\delta}+2\boldsymbol{t})\boldsymbol{N}+\boldsymbol{t}(2\boldsymbol{M}-1))}{2\boldsymbol{N}}$.

(2) In MM, social welfare is independent of the unit donation d. In MN, if $0 < d < \breve{d}\left(d > \breve{d}\right)$, *social welfare increases (decreases) with d, where*

$$\breve{d} = \frac{(1-r)\left(\delta(2r-1)MNT + 2(r-1)\left(2t\left(M^2-N^2\right) - \delta\left(M^2+N^2\right) + t(N-M)\right)\right)}{((2r-3)N - 2r + 2)M^2 + \left((2r-3)N^2 + N\right)M + 2N^2(1-r)} \text{ and } \widetilde{d} > \breve{d}$$

Proposition 4 explores the impact of unit donation amounts on total donation and social welfare. Intuitively, a larger unit donation would logically lead to higher total donations since they constitute the overall donation. However, Proposition 4**(1)** intriguingly reveals that when the

unit donation is substantial, total donations actually decrease as the unit donation increases. This counterintuitive finding can be explained as follows: in a competitive market between CM and NCM sellers, indiscriminately raising unit donations often results in price hikes. If the price of CM products exceeds what consumers are willing to pay, those considering CM products due to their prosocial appeal may opt for NCM products due to the significant price gap. Empirical research by Koschate-Fischer, Diamantopoulos, and Oldenkotte [15] corroborates this, demonstrating that excessively high unit donations reduce consumer willingness to pay. Consequently, when consumer prosociality is relatively low and unit donations are high, the demand for CM products decreases, leading to a reduction in total donations.

Similarly, Proposition 4**(2)** indicates that an increase in unit donation does not necessarily translate into higher social welfare when CM and NCM sellers fully compete. Social welfare is composed of seller and platform profits, consumer surplus, and total donation. Let's see how these factors change with the unit donation d. Firstly, for CM sellers, when $0 < d < \frac{(2tM+N\delta+2Nt-t)(1-r)}{N} = d_i$, we have $\frac{\partial \pi_i}{\partial d} = \frac{2((2(r-1)t+r\delta+d-\delta)N+t(2M-1)(r-1))N}{(2T-1)^2(1-r)tT} < 0$. After comparing d_i and $\breve{d}$, we find $0 < \breve{d} < d_i$. Therefore, we conclude that the profit of CM sellers decreases with the unit donation d when $d < \breve{d}$. Secondly, for NCM sellers, when $0 < d < \frac{(M\delta-2tM-2Nt+t)(1-r)}{M} = d_z$, we have $\frac{\partial \pi_z}{\partial d} = \frac{2(((2-2r)t+r\delta+d-\delta)M-t(2N-1)(r-1))M}{(2T-1)^2(1-r)tT} < 0$. Also, after comparing d_z and $\breve{d}$, we find $0 < d_z < \breve{d}$ in this case. Thus, the profit of NCM sellers increases with the unit donation d when $d < \breve{d}$. Using the similar approach, we find that the platform's profit increases with the unit donation, while consumer surplus decreases with the unit donation. As the unit donation amount increases, the price of CM products will also rise. This leads to a reduction in consumer utility and demand, which subsequently affects the profitability of CM sellers. When the decreases in the profit of CM sellers and consumer surplus are more significant than the increases in the profits of NCM sellers, the platform, and the total donation, social welfare decreases with the unit donation. Therefore, considering the trade-off between these factors, social welfare initially increases with the unit donation but eventually decreases.

If we take a closer look at the two thresholds $\widetilde{d}$ and $\breve{d}$ given by Proposition 4, we find $\widetilde{d} > \breve{d}$ consistently applies. The total donation amount keeps increasing with the unit donation when $d < \breve{d}$. Although the profit of the platform and NCM increases with the unit donation d, the profit of CM sellers and consumers surplus decreases with unit donation d. Therefore, we get $\widetilde{d} > \breve{d}$. This reflects a higher unit donation can potentially bolster the platform's profitability, but not necessarily lead to a higher social welfare.

Fig. 7 illustrates how total donations and social welfare vary with unit donation in MN when consumer prosociality $\delta = 3.5$. These figures show data for a market with 60 sellers, including 35 CM sellers, a commission rate $r = 0.08$, and a transportation cost factor $t = 1$. They highlight the importance of optimizing unit donation in highly competitive scenarios to achieve both social and economic goals. By comparing the figures below, it is observed that in the MN scenario, the optimal value for social welfare appears earlier than the optimal value for total donations as the unit donation amount increases. This also indicates that a higher unit donation doesn't necessarily lead to a higher social welfare.

From Proposition 4, we derive several key management insights. Firstly, Proposition 4 explains why platforms like Taobao have reduced fixed donation standards for charitable products, i.e., goods for good. Secondly, it advises sellers against blindly pursuing high unit donations in cause marketing (CM). Thirdly, it suggests that platforms should set reasonable rules for CM, adjusting donation standards to improve total donations effectively. Platforms can also guide sellers in determining optimal donation amounts based on these standards. Fourthly, it emphasizes the need for platforms to balance social and economic benefits when setting unit donation levels. These insights help sellers and platforms navigate the complexities of CM, ensuring strategies that enhance both profitability and social impact.

5. Extension

5.1. Price rate

The base model considers donations made as a fixed unit amount. However, in practice, donations can also be made as a fixed percentage of the price. For instance, in Taobao's *Goods for Good* program, sellers have the option to donate either a fixed unit amount or a fixed percentage of the price. Similarly, on the Amazon shopping platform, 0.5% of the sale price of items bought from the charity section is donated to charitable organizations.

This section extends the model to include donations made as a percentage of the retail price (price rate). When donations are made as a fixed percentage, the profit expression for NCM sellers remains the same as in Eq. (4). However, the profit expression for CM sellers and the total donation expression are given by:

$$\Pi_i^{CM} = [(1-r)(1-\epsilon)p_i]Q_i^{CM}, i = 1, 2, \ldots, m, \tag{17}$$

$$\omega = \sum_{i=1}^{m} \epsilon p_i Q_i^{CM}, \tag{18}$$

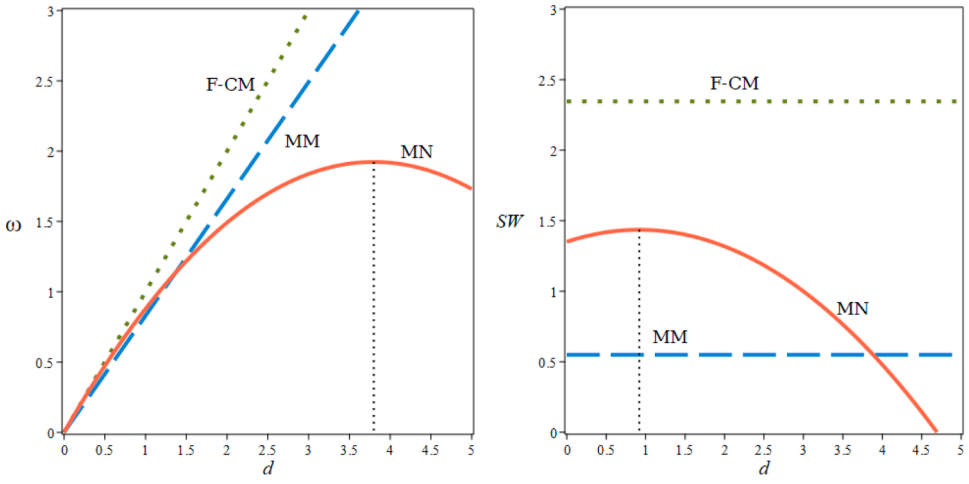


Fig. 7. The impact of unit donation on total donation and social welfare.

where ϵ represents the price rate when the donation is made as a fixed percentage of the retail price.

The calculation and equilibrium outcomes are shown in **Appendix A**.

Lemma 4. *When the donation is made as a price rate, sellers' profits in F-CM are lower than in N—CM.*

Lemma 4 compares the profit levels of sellers under two special cases when donations are made as a fixed percentage of the product's retail price. The underlying reason for this result is that when donations are given as a price rate, sellers cannot transfer the donation costs by raising product prices. The magnitude of the price rate ϵ directly impacts the sellers' profit levels. However, as is shown in Lemma 2, when donations are made as a fixed amount d, sellers can pass on the donation costs to consumers by increasing the unit price of the product. A comparison reveals that donating a fixed amount is a more optimal choice for sellers. This also corroborates the reality that a far greater number of sellers on the same platform choose to donate with fixed amount rather than using price rate.

Proposition 5. *When the donation is given by price rate, there exists a threshold $\bar{\epsilon} \in (0,1)$ such that it is more profitable for the seller to implement CM (i.e. $\pi_i^* > \pi_z^*$) if and only if $\epsilon < \bar{\epsilon}$, where* $\bar{\epsilon} = \dfrac{\delta T\left(\left(T-\frac{1}{2}\right)t-\frac{1}{4}\delta(M-N)\right)}{\left(\left(T-\frac{1}{2}\right)t+\frac{1}{2}N\delta\right)^2}$.

Proposition 5 highlights that the price rate significantly shapes sellers' strategies in cause marketing when donations are based on a percentage of the product price. When the price rate is relatively low, sellers leverage the warm glow effect to stimulate additional positive utility among consumers through cause-related products. This approach boosts product sales and consequently improves profitability. However, as the price rate rises, sellers must carefully manage the trade-off between the increased demand spurred by cause marketing and the accompanying donation expenses. Excessively high price rate can undermine sellers' profitability.

The managerial implications of Proposition 5 are as follows: Firstly, sellers should meticulously determine the price rate when participating in cause marketing, especially for high-priced products, favoring lower price rate. Secondly, this proposition underscores the rationale behind initiatives such as Alibaba's charity campaign, which officially recommends donations from *goods for good* products to be set at only 0.3% of retail prices. This indicates that, from the platform's standpoint, maximizing total donations does not necessarily rely on high price rates.

Proposition 6. *In MM, the total donation in P-CM is always higher than F-CM; In MN, when $0 < \delta < \widetilde{\delta}$ ($\widetilde{\delta} < \delta < \overline{\delta}$), the total donation in P-CM is lower (higher) than F-CM, where* $\widetilde{\delta} = \dfrac{\left(\sqrt{M^2+MN}-M\right)(2T-1)t}{MN}$, $\overline{\delta} = \dfrac{2T-1}{T-1}$.

Proposition 6 indicates that increasing the number of CM sellers does not necessarily lead to a higher total donation. When there is intense competition between CM and NCM sellers and consumers have high prosociality, having only some sellers opt for cause marketing may result in higher total donations than if all sellers engage in cause marketing. This is because consumers with higher prosociality are willing to bear more donation costs to support charitable causes.

Furthermore, when donations are made as a price rate, the pricing of CM products directly influences the amount of donations. However, when consumers have low prosociality, high priced CM products may reduce demand, thereby lowering total donations. This illustrates the complex relationship between CM product pricing and consumer behavior, necessitating consideration of market competition, consumer preferences, and pricing strategies.

Fig. 8 illustrates the variations in total donations under scenarios where CM sellers monopolize the market (MM), CM and NCM sellers compete fully (MN), and all sellers opt for cause marketing (F-CM) when warm glow effect $\delta = 0.5,\ 2.5$ respectively. From Fig. 8, it is observed that when consumers have low prosociality, total donations under F-CM exceed those under MN; conversely, when consumers have high prosociality, total donations under F-CM are less than those under MN.

When donations are made in the form of a fixed unit donation amount, the conclusion of Proposition 6 does not hold. In this scenario, sellers can pass the cost of donations onto consumers, and there is no direct relationship between the product price and the unit donation amount. This can be explained by sellers potentially setting higher prices due to the warm-glow effect, without a corresponding increase in the unit donation amount. For example, sellers might exploit consumers' warm-glow effect by charging prices higher than the actual cost of the donation to capture more consumer surplus. Therefore, even when the price of cause-marketing products is higher and consumers' prosociality is greater, donating based on a fixed unit amount does not necessarily lead to a higher total donation amount. This highlights the necessity to consider the impact of donation format, pricing strategy, and consumer behavior when analyzing donation behavior.

We can conclude following managerial insights from Proposition 6. When many sellers choose to donate at a fixed rate of the price and consumers have high prosociality, the platform can implement several measures to manage these sellers. Firstly, the platform can set qualification criteria for sellers, such as requiring a certain level of historical sales, store credit rating, and product quality. By setting these thresholds, the platform can control the number of CM sellers, preventing all sellers from engaging in cause marketing and thereby increasing the overall donation amount. Additionally, these measures can improve the overall quality of cause-marketing sellers, creating a positive platform environment.

Jiang and Zou [47] reach similar conclusions but mainly focus on the impact of consumer search costs on platform profitability, considering different commission fee models. This paper extends their work by examining the impact of consumers' warm-glow effect on total donations under different donation models. This extension provides a more comprehensive understanding of platform management and aids in

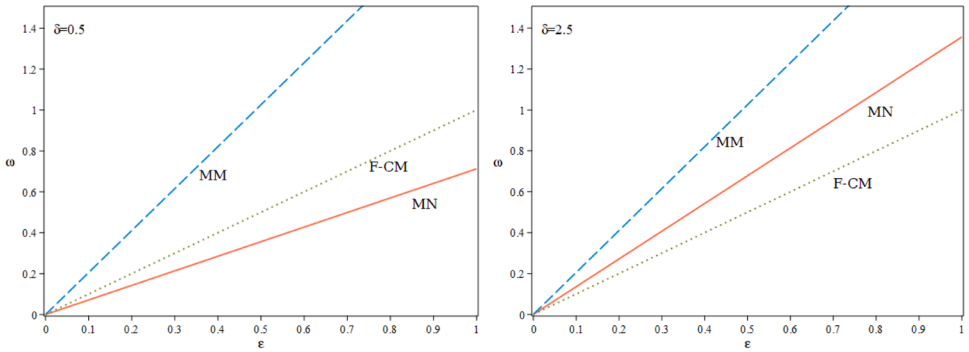


Fig. 8. The impact of price rate on total donation.

developing more effective management strategies.

5.2. Different commission fee rate

In our discussion of the implications of cause marketing (CM) on e-commerce platforms, we examine how varying commission fees can influence a seller's optimal CM decision. Specifically, this section analyzes the robustness of our main findings by considering different commission fee rates set by the platform.

From Proposition 3, we know profit of CM sellers consistently exceeds that of NCM sellers under various commission fee structures in the MM scenario. And in the MN scenario, where CM and NCM sellers are in full competition, when consumers' prosociality is low and the number of CM sellers is small, the seller should adopt CM. Conversely, when consumers' prosociality is low and there are many sellers adopting CM, the seller should not adopt CM. Fig. 9 illustrates the optimal decision for sellers regarding the adoption of the CM strategy in the MN scenario based on different commission fee rate levels ($r = 0.1, 0.2, 0.3, 0.4$) imposed by the platform, where we assume the number of NCM seller $N = 25$, consumers warm-glow effect $\delta = 6$, the fixed unit donation $d = 5$, and the consumer's sensitivity to product characteristics $t = 1$. From above four subfigures, it can be observed that when consumer prosociality is relatively low and the number of CM sellers is relatively low, the seller is more inclined to adopt the CM strategy to achieve higher profitability. To encourage more third-party sellers to join the CM strategy, the platform should not charge a high commission fee to sellers. Furthermore, our numerical analysis confirms the robustness of these results.

6. Conclusion

Cause marketing integrates transactions and donations on e-commerce platforms, effectively linking platforms, third-party sellers, nonprofit organizations, and consumers. However, not all third-party sellers on these platforms adopt cause marketing strategies. Building upon the spokes model, this paper constructs a horizontal differentiation competition model with numerous cause-marketing sellers and non-cause-marketing sellers. By introducing consumers' warm-glow effect, we consider the impact of consumer prosociality, unit donation amount, and the number of cause-marketing sellers on pricing, profitability, and total donation for both the platform and third-party sellers. The base model is then expanded to include scenarios where donation is given by a percentage of price. Moreover, we use a numerical analysis of different levels of commission fee rates to demonstrate the robustness of our main results.

Through the exploration of optimal cause-marketing strategies on e-commerce platforms, this paper offers the following managerial insights.

For platforms, firstly, they can control options for fixed unit donation amounts or fixed donation percentages at lower levels to maintain demand for cause-marketing sellers, leveraging the advantage of "small amounts from a large volume" in transactions(Proposition 4 & Proposition 5). Secondly, platforms can set thresholds to regulate the number of cause-marketing sellers when many sellers choose fixed percentage price donations and consumer prosociality is high, thus increasing overall donation amounts(Proposition 6). Thirdly, platforms should regularly update consumers on their purchases of cause-marketing products and implement reward measures like virtual certificates or loyalty points to enhance consumer prosociality and encourage participation in cause-marketing activities(Proposition 1).

For sellers, we identify the conditions under which sellers should adopt CM when there are multiple competitors selling the homogeneous products in the platform(Proposition 3). We reveal that the decision to adopt CM is influenced by the choices of other sellers and the prosociality level of consumers. Specifically, they should consider adopting CM when most sellers are using it. The dynamics change when CM and NCM sellers compete fully. When consumers' prosociality is relatively low and the number of CM sellers is small, the seller should adopt cause marketing; however, when the number of CM sellers is large, this strategy is less effective. Conversely, when consumers' prosociality is relatively high, adopting cause marketing is advisable regardless of the number of CM sellers. Therefore, before engaging in cause marketing, sellers should investigate consumers' attitudes toward CM, which may

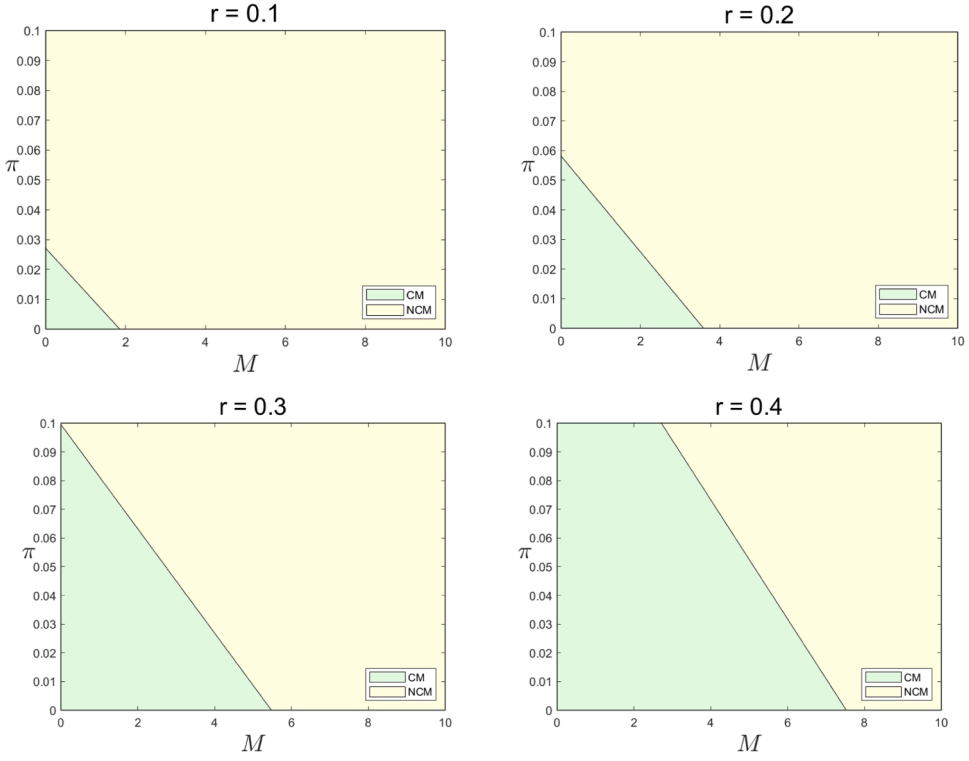


Fig. 9. The impact of the number of CM seller on optimal CM decision with different r.

be reflected in their level of prosociality. Additionally, sellers need to be aware of the number of competitors offering similar products with cause marketing. Secondly, sellers should prioritize fixed unit donations, especially when unit prices are high, and carefully consider fixed percentage donations(Lemma 4). Thirdly, when opting for fixed percentage donations, sellers should set lower rates to attract more demand and increase profits(Proposition 5).

This paper offers several key theoretical contributions to the field of cause marketing. Firstly, we construct a novel spokes model that incorporates the warm-glow effect, broadening the scope of research on cause marketing by explicitly considering consumer psychological responses. Secondly, we define and analyze the concept of actual donations by cause-marketing sellers on e-commerce platforms, bridging a gap in the literature by formalizing this critical aspect of online cause marketing. Thirdly, Our analysis reveals that increasing either unit donation amounts or the number of cause-marketing sellers does not necessarily lead to higher total donations or improved social welfare, challenging conventional assumptions and providing new perspectives on cause-marketing. Fourthly, by comparing fixed donation amounts with fixed percentage-based donations, we provide a theoretical foundation for optimal donation strategies, aiding both platforms and sellers in making informed cause-marketing decisions. Fifthly, we demonstrate the rationality of certain platform-level policies, such as Taobao's decision to lower unit donation standards and recommend low price rates for sellers, offering practical insights for policymakers and practitioners.

The limitations and future directions of this paper is following. Firstly, the model assumes equal likelihood $(1/T)$ of selection for cause-marketing and non-cause-marketing sellers. Future research could consider how product sorting and consumer search costs influence consumer choice, potentially favoring higher visibility and selection rates for cause-marketing products. Secondly, we assume each seller offers only one product in a horizontally differentiated market. Future studies could explore cause-marketing strategies for sellers offering multiple products with varying quality in vertically differentiated markets. Thirdly, the model depicts consumers choosing between two sellers, whereas consumers often consider multiple options before making a purchase. Future research could expand the choice set to $2 \leq k \leq T$, reflecting more realistic consumer behavior. By delving into optimal cause-marketing strategies on e-commerce platforms, this study provides theoretical insights and practical guidance for enhancing platform and seller decision-making in cause marketing.

CRediT authorship contribution statement

Qian Su: Writing – review & editing, Writing – original draft, Software, Project administration, Methodology, Formal analysis, Conceptualization. **Xiaoyang Zhou:** Writing – review & editing, Supervision, Resources, Methodology, Conceptualization. **Qi Liu:** Writing – review & editing, Supervision, Resources, Methodology, Funding acquisition. **Kai Zhang:** Writing – review & editing, Supervision, Methodology, Conceptualization.

Declaration of competing interest

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work, there is no professional or other personal interest of any nature or kind in any product, service and/or company that could be construed as influencing the position presented in, or the review of, the manuscript entitled, "Optimal Cause Marketing Strategies for Online Platforms and Third-Party Sellers: A Spokes Model Analysis".

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.omega.2024.103271.

Data availability

No data was used for the research described in the article.

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